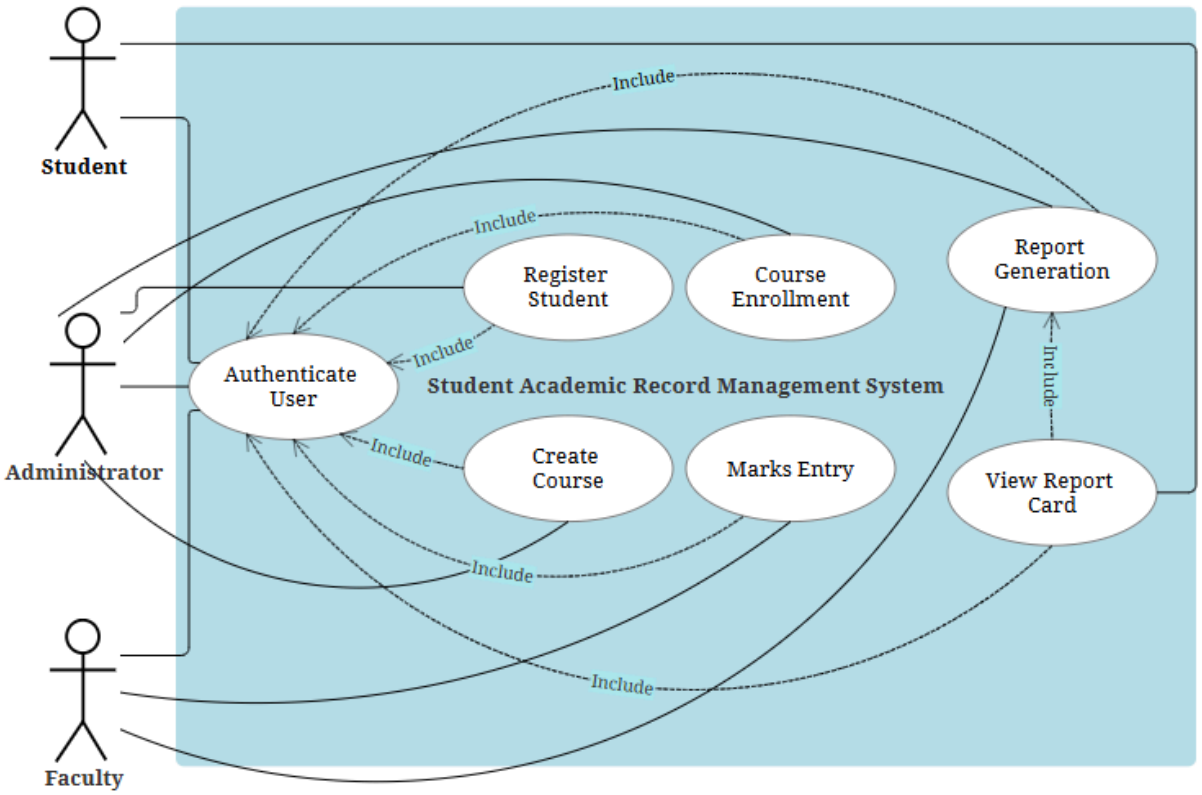
Student Academic Record Management System (SARMS)

Use Case Document

Group 9

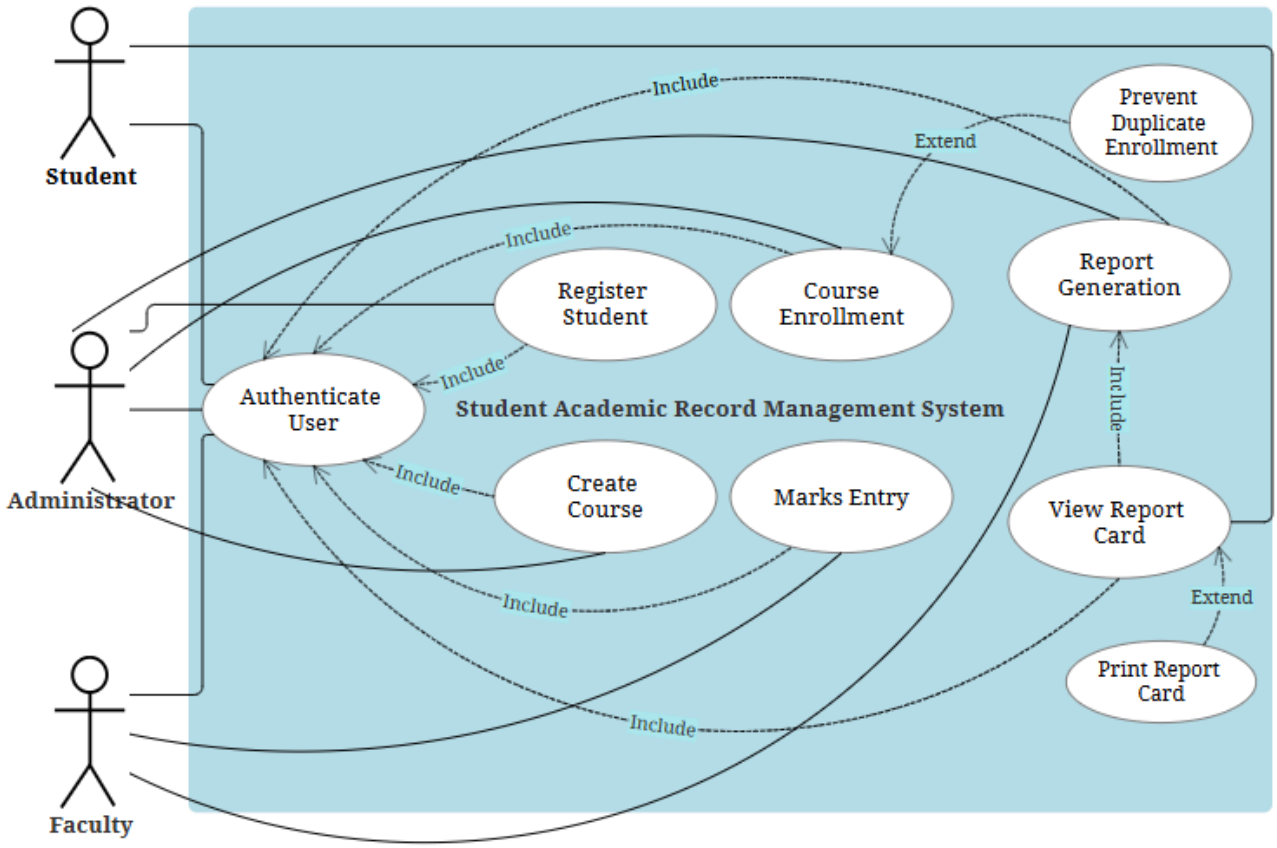
Use Case Diagram

1. Use Case Diagram of SARMS



This use case diagram shows how the main users, Administrator, Faculty, and Student, interact with the SARMS system. It includes key functions like registration, course management, marks entry, and report generation. The <<include>> relationships represent mandatory actions such as authentication and required system steps.

2. Extended Use Case Diagram of SARMS



This extended use case diagram shows both mandatory and optional system behaviours. The <<include>> relationships represent required actions like authentication, while the <<extend>> relationships show optional features such as preventing duplicate enrollment and printing the report. It gives a clearer picture of how the system handles both core functions and extra conditional actions.

Use Case Document

Process - 0.1: Student Registration -

ID:	UC-0.1
Title:	Student Registration
Description:	This use case allows the administrator to register a new student into the system by entering personal and academic details.
Primary Actor:	Administrator
Secondary Actor:	Authentication System
Preconditions:	● The administrator must be authenticated. ● Students should not already exist in the system.
Postconditions:	● Student records are stored in the database. ● A unique roll number is generated.
Dependency	«include» Authenticate User
Generalization	None (There is no parent or child use case for Student Registration.)
Main Success Scenario:	● Administrator selects “Register Student”. ● System prompts for student personal and academic details. ● Administrator enters required details. ● The system validates the entered data. ● The system generates a unique roll number. ● The system stores the student record in the database. ● A confirmation message is displayed.
Extensions or Alternate Flow:	**Invalid Data Entered:** ○ If any required field is missing or invalid, The system displays an error message.

	Administrator re-enters correct information. **Student Already Exists:** ○ If student record already exists, The system displays a duplication warning. The registration process is cancelled.
Frequency of Use:	High (Student registration happens every admission cycle.)
Status:	Implemented
Owner:	Group 9
Priority:	High (Student registration is a core system functionality.)

Process - 0.2: Course Creation -

ID:	UC-0.2
Title:	Course Creation
Description:	Allows administrators to create new course information including credits and syllabus.
Primary Actor:	Administrator
Secondary Actor:	None
Preconditions:	Administrator authenticated.
Postconditions:	Course record saved in database.
Dependency	«include» Authenticate User

Generalization	None
Main Success Scenario:	• Administrator enters course details. • The system validates uniqueness. • System assigns course code. • Course stored in database. • Confirmation displayed.
Extensions or Alternate Flow:	If duplicate course → error message.
Frequency of Use:	Medium
Status:	Implemented
Owner:	Group 9
Priority:	High

Process - 0.3: Enroll Student -

ID:	UC-0.3
Title:	Enroll Student
Description:	Allows administrators to enroll students into selected courses.
Primary Actor:	Administrator
Secondary Actor:	None
Preconditions:	• Students registered. • The course exists.

Postconditions:	Enrollment record stored.
Dependency	«include» Authenticate User
Generalization	None
Main Success Scenario:	• Administrator enters roll number. • The system retrieves student records. • The administrator selects courses. • The system validates enrollment. • Enrollment stored in database. • Confirmation displayed.
Extensions or Alternate Flow:	If invalid course → error message. If already enrolled → prevent duplication.
Frequency of Use:	Medium
Status:	Implemented
Owner:	Group 9
Priority:	High

Process - 0.4: Enter marks -

ID:	UC-0.4
Title:	Enter Marks
Description:	Allows faculty to enter marks of enrolled students.
Primary Actor:	Faculty

Secondary Actor:	Academic Record System
Preconditions:	• Faculty logged in. • Students enrolled.
Postconditions:	Marks stored and SWA triggered.
Dependency	«include» Calculate SWA
Generalization	None
Main Success Scenario:	• Faculty selects courses. • Faculty enters marks. • The system validates marks. • The system calculates grades. • Marks stored. • SWA calculation triggered.
Extensions or Alternate Flow:	Marks out of range → error message.
Frequency of Use:	High
Status:	Implemented
Owner:	Group 9
Priority:	High

<u>Process - 0.5: Calculate SWA & CSWA -</u>

ID:	UC-0.5
Title:	Calculate SWA & CSWA

Description:	Automatically calculates semester and cumulative weighted averages.
Primary Actor:	System
Secondary Actor:	None
Preconditions:	Marks and credit data available.
Postconditions:	SWA, CSWA and standing updated.
Dependency	None
Generalization	None
Main Success Scenario:	• The system retrieves marks. • The system retrieves credits. • The system calculates SWA. • The system calculates CSWA. • The system determines academic standing. • Results stored in the database.
Extensions or Alternate Flow:	Missing marks → calculation aborted.
Frequency of Use:	Automatic
Status:	Implemented
Owner:	Group 9
Priority:	High

<u>Process - 0.6: Generate Report Card -</u>

ID:	UC-0.6
Title:	Generate Report Card
Description:	Allows students to view report cards with SWA, CSWA, and academic standing.
Primary Actor:	Student
Secondary Actor:	Report Generation Module
Preconditions:	Academic results available.
Postconditions:	Report card displayed.
Dependency	«include» Calculate CSWA «include» Determine Academic Standing
Generalization	None
Main Success Scenario:	• Student requests report. • The system retrieves academic data. • System formats report. • Report displayed.
Extensions or Alternate Flow:	If results unavailable → notification displayed.
Frequency of Use:	Medium
Status:	Implemented
Owner:	Group 9
Priority:	High